

Every missense substitution
in MLH1, MSH2, MSH6, PMS2



Features
gnomAD v4 frequency
AlphaMissense
AlphaFold structure
genomic position

Labels
ClinVar ≥ 2 stars
(expert-panel subset)
ProteinGym DMS
(one MSH2 assay)

Axis A: feature sources
(ClinVar labels)
A0 sequence window only
A1 gnomAD A2 AlphaMissense
A3 structure A4 genomic
M gnomAD + AlphaMissense
A5 all sources

Axis B: label sources
(all features)
A5 ClinVar
B3 ClinVar expert panel only
B1 MSH2 DMS only
B2 ClinVar + MSH2 DMS

7 models $\times$ 3 seeds

GBM, MLP,
window MLP, CNN,
BiLSTM,
BiLSTM + attention,
Transformer

Leave one
gene out

scored on the
held-out gene's
ClinVar labels

threshold from
inner validation
(training genes)

AUROC, MCC,
paired bootstrap